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Ethio-Parents' Schools, Addis Ababa/Hawassa



Second Semester Centralized Biology Final Exam for Grade- 9
June, 2020

Time Allowed: 70 Minutes

General Directions

1. Use legible/clear handwriting
2. For your answers use the separate answer sheet provided.
3. Use uppercase letters for your answers (for Parts I through V).
4. Never use pencils to write your answers. Corrections are forbidden for any work done with a pencil.
5. Make sure that this exam consists of EIGHT printed pages.

Part I. Write 'TRUE' if the statement is correct or 'FALSE' if it is incorrect and write down your answer on the separate answer sheet provided.

1. Bryophytes have conducting tissue.
2. Living things are classified and named to avoid confusion.
3. Open circulatory system is the characteristics of mammals.
4. Gymnosperms differ from Angiosperms in having covered seed.
5. T-cells make antibodies which bind to the antigen and destroy it.
6. The most diverse and adapted groups of plants are angiosperms.
7. The binomial system of nomenclature was developed by Robert Koch.
8. One of the biggest problems with HI V/AIDS is the inadequacy of drugs.
9. Antiseptics are chemical that are applied to living tissue to kill micro-organisms.
10. The simplest and best known method of sterilizing thing using heat involves boiling.

Part II. Match the premises under the column "A" with the responses given under the column "B" by writing the letter of your choice on the separate answer sheet provided. All the words may not be used.

Column 'A'

11. Sexually transmitted infections (STIs)
12. It is caused by *Haemophylus ducreyi*
13. Men infected with it will have a burning sensation
14. It occurs in stages called primary, secondary and late

Column 'B'

- A. Syphilis
- B. Gonorrhea
- C. Chancroid
- D. Trichomoniasis
- E. Venereal diseases
- F. Deficiency diseases

Part III. Fill in the blank spaces with the most appropriate words on the separate answer sheet provided. Use the words inside the box below. All of the words may not be used.

- | | | |
|-------------------|------------------|----------------|
| ♦ Prokaryotic | ♦ Autoclaving | ♦ Incineration |
| ♦ Dichotomous key | ♦ Pasteurization | ♦ Lymphocytes |
| ♦ Animalia | ♦ Camouflage | ♦ Echinoderm |
| ♦ Amphibian | ♦ Morphology | |

15. Humans are a member of the kingdom _____.
16. The form or appearance of an organism is called _____.
17. The simplest type of identification key is called the _____.
18. An animal that have tube feet for locomotion and feeding is known as _____.
19. The white blood cells forming antibodies against microbes are known as _____.
20. Animals that spend part of their lives in water and part of it on land are called _____.

21. The cells that do not have a true nucleus or membrane-bound organelles are said to be _____.
22. The ability of an animal to blend its color into its surroundings to avoid detection IS known as _____.
23. The burning of substances at high temperatures in the air which also kills micro-organisms is called _____.
24. A flexible rod-like structure, made of cartilage, which runs along the dorsal side of the body is known as _____.

Part IV. Choose the correct answer from the alternatives given and write the letter of your choice on the separate answer sheet provided.

25. Asexual reproduction in yeast cells is carried out by
 - A. budding.
 - B. binary fission.
 - C. fragmentation.
 - D. multiple fission.
26. Bacteria are classified by which of the following shapes?
 - A. Rod, cone, spiral
 - B. Rod, spherical, cone
 - C. Rod, spherical, spiral
 - D. Spherical, cone, spiral
27. The Germ theory of disease states that
 - A. all organisms arise from non-living matter.
 - B. all organisms come from pre-existing organisms.
 - C. infectious diseases are caused by microorganisms.
 - D. infectious diseases are caused by macroorganisms.
28. Which of the following is a risk of overusing antibiotics to treat infections? Antibiotics may
 - A. increase the rate of infection.
 - B. decrease the rate of infection.
 - C. increase bacterial resistance to the antibiotics.
 - D. decrease bacterial resistance to the antibiotics.
29. Malaria parasite reproduces inside the
 - A. epithelial cells.
 - B. red blood cells.
 - C. white blood cells.
 - D. reproductive cells.

30. A group of organisms that can interbreed successfully to produce a fertile offspring is the characteristic of members of a single
- A. order.
 - B. family.
 - C. phylum.
 - D. species.
31. One of the following may NOT be a reason to classify organisms systematically. To
- A. simplify their study.
 - B. study organisms that only benefit us.
 - C. study the relationship between organisms.
 - D. put them in groups based on their similarities and differences.
32. The modern classification of living things is introduced by
- A. Aristotle.
 - B. Carl Linnaeus.
 - C. Robert Hooke.
 - D. John Needham.
33. Which one of the following taxonomic groups is the most complex hierarchy in the classification of organisms? The
- A. phylum.
 - B. genera.
 - C. division.
 - D. kingdom.
34. Suppose your friend tells you that he suspects he is HIV positive. How would you react to this? I will advise him to
- A. buy and take anti-retroviral drugs.
 - B. avoid his friendship to protect others.
 - C. go to health center to know his status.
 - D. inform other friends that he is HIV positive.
35. In the following table which alternative holds the correct comparison between T-cells and B-cells of the human immune system? They

	T-cells	B-cells
A.	are types of red cells.	are types of platelets.
B.	make lots of 'killer T-cells'.	make memory cells.
C.	targeted white blood cells.	targeted red blood cells.
D.	make antibodies which bind to antigens.	bind to antigens and destroy them.

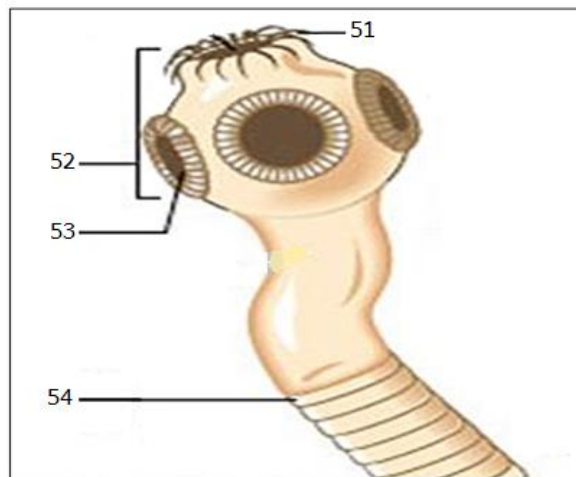
36. What do you call the treatment for HIV/AIDS?
- A. Mantoux skin test
 - B. Intravenous fluids
 - C. Chest and back x-ray
 - D. Antiretroviral therapy (ART)

37. Which medical service gives people advice before and after they have HIV test?
- A. Family planning and spacing
 - B. People living with HIV-AIDS (PLWHA)
 - C. Voluntary counseling and testing (VCT)
 - D. Highly active antiretroviral therapy (HAART)
38. In Gram's staining technique, Gram's iodine is used as a
- A. mordant.
 - B. decolorizer.
 - C. counter stain.
 - D. primary stain.
39. Which traditional medicine in Ethiopia is used to treat tapeworm?
- A. 'Feto'
 - B. 'Koso'
 - C. 'Tenadam'
 - D. 'Dingetegna'
40. The biggest flowers in the world belong to
- A. *Zea mays*.
 - B. *Wolffia globosa*.
 - C. *Rafflesia arnoldii*.
 - D. *Lantana camara*.
41. Which group of plants belong to spermatophytes?
- A. Bryophytes
 - B. Filicinophytes
 - C. Pteridophytes
 - D. Coniferophytes
42. Which one of the following animal is an example of cnidarian?
- A. Hydra
 - B. Planaria
 - C. Tape Worm
 - D. Hook worm
43. Which animals are dorso – ventrally flattened?
- A. Annelids
 - B. Nematodes
 - C. Coelenterates
 - D. Platyhelminthes
44. To which group of mammals do we human beings belong? The
- A. placental
 - B. marsupials
 - C. monotremes
 - D. pouched mammals
45. Which of the following is different from the rest of organisms?
- A. Spider
 - B. Locust
 - C. Butterfly
 - D. Housefly
46. Denitrifying bacteria use
- A. nitrites as energy source and break them down into nitrogen gas.
 - B. nitrates as energy source and break them down into nitrogen gas.
 - C. ammonium compounds as energy source and break them down into nitrate.
 - D. ammonium compounds as energy source and break them down into nitrogen gas.

47. Most plants that live in dry conditions have
A. large fleshy leaves with a thick cuticle.
B. increased the surface area of their leaves.
C. curled leaves to reduce the surface area of the leaf.
D. curled leaves to increase the surface area of the leaf.
48. In a simple food chain the primary consumers are found at
A. 1st trophic level.
B. 2nd trophic level.
C. 3rd trophic level.
D. 4th trophic level.
49. Which graphical representation shows the feeding relationship and the number of organisms at each trophic level? Pyramid of
A. life
B. energy
C. biomass
D. numbers
50. Which part of the carbon cycle occurs when plants, trees, or fossil fuels are burned?
A. Decay
B. Respiration
C. Combustion
D. Photosynthesis

Part V. The diagram below illustrates the structure of a tapeworm. Identify the parts designated by the numbers 51-54 and write down your answer on the separate answer sheet provided. Use the alternative words inside the box below. All words may not be used.

- | | | |
|--------------|----------|--------------|
| • Cysticerci | • Scolex | • Proglottid |
| • Neck | • Sucker | • Hook |



Part VI. Read the following questions carefully and write down brief answers on the separate answer sheet provided.

- 55. What are homologous structures?
- 56. What is the principle of autoclave?
- 57. What do you understand by mycorrhizae?
- 58. What are the simple rules for writing scientific names for living organisms?
- 59. Mention the different lymph nodes. Where are their largest groupings found in your body?
- 60. Suppose a student stepped on a rusty nail. What is the disease risk factor? How could it be controlled?